

Week 1: Probability, Distributions, and Model Comparison

DSAA1001 · Three-tier learning sheet · Advanced Learning Sheet

Opening diagnostic challenge. A campus lunch operations team must decide before 11:30 how many meals, couriers, and pickup counters to prepare. The forecast says rain or no rain; a promotion may be on or off; the app gives an early, noisy demand signal. Later we will observe true demand, late deliveries, delay times, and complaints.

Before a statement like “30% chance of high demand” can guide staffing, what must the team write down? Name the possible situations or quantities that matter, and say what would make the 30% number meaningful.

Reading contract and capabilities

Week orientation

Backbone question. How do we represent uncertainty with quantities that we can calculate, update after observing information, simulate, and compare with data?

Running example. A campus lunch operations team chooses staffing, meal preparation, and courier allocation before the true rush is known. They observe weather forecast W , promotion status K , early app signal Y , and later outcomes: true demand D , lateness count L , delay time T , and complaints C . They compare candidate demand models P and Q before deploying decisions.

A **distribution** is a table, curve, or rule that tells us how probability is assigned to values.

Learning objectives

- construct and critique probability spaces;
- distinguish outcomes, events, random variables, values, and distributions;
- read PMFs, PDFs, CDFs, joint laws, marginal laws, and conditional laws;
- use conditioning, product rules, sum rules, and Bayes' rule;
- transform random variables and simulate by inverse transform;
- manipulate Gaussian models, sums, expectations, and variances;
- use entropy and KL divergence to compare predictive models and inform decisions.

Breadcrumb Prompt

Write your starting rule of thumb

In one sentence, write how you currently handle an uncertain decision such as staffing for lunch. At the end of the sheet, revise it into a sequence of modeling steps.

- **Tier 1: fast path.** Operational ideas you should be able to use immediately.
- **Tier 2: formal machinery.** Definitions, theorems, and formulas that make the fast path reliable.
- **Tier 3: extension and application.** Construct, compare, and apply Tier 2 ideas in less routine problems.

Pause and Think

Question: When you first see a problem statement that says “random” or “uncertain,” what is your default approach?

Hint: For the lunch team example, would you start by listing possible order counts, or would you immediately reach for a probability distribution formula?

Answer:

A robust approach starts by defining what could happen and what is observed. Formulas come after the modeling objects are clear.

1. Probability spaces and granularity

A probability model is only useful after we decide what the model is allowed to talk about. For the lunch team, the first move is not a formula; it is deciding which possible day/order states must be distinguished for staffing, stockout, and delivery decisions.

Tier 1: the model begins with outcomes

A probability model has three types of ingredients:

- **outcomes** (individual elements): the mutually exclusive possible results; we write Ω for the set of all outcomes—this complete list is called the **sample space** (think of it as the answer to “what could happen?”);
- **events** (sets): collections of outcomes that represent questions; for example, $\{\text{late}\}$ represents “is the order late?”;
- **probabilities** (a function): a rule P that assigns a number between 0 and 1 to each event; we can also write $P(\omega)$ for the probability of a single outcome ω , which formally means $P(\{\omega\})$. We write $\{\omega\}$ with curly braces because P technically assigns probability to sets (events), not individual outcomes directly.

For one order, $\Omega = \{\text{late, on time}\}$ may be enough if we only care about punctuality. If weather matters, refine to $\Omega = \{(\text{late, rain}), (\text{on time, rain}), (\text{late, dry}), (\text{on time, dry})\}$. An event like $A = \{(\text{late, rain}), (\text{late, dry})\}$ represents “the order is late, regardless of weather.” If delay time matters, use an outcome that contains a time value, or define a random variable (introduced shortly) on top of a richer outcome space.

The sample space should be **collectively exhaustive** (nothing is missing), **mutually exclusive** (outcomes can’t overlap), and at the **right granularity** (appropriate detail level) for the decision. For example, in a coin toss: collectively exhaustive means listing both H and T; mutually exclusive means a single toss cannot be both H and T; right granularity means we don’t need the coin’s exact rotation angle.

Sample-space warning. For a coin toss, $\Omega = \{H, T, HT\}$ mixes one-toss and two-toss outcomes, so outcomes are not mutually exclusive at one granularity. $\Omega = \{H\}$ is not exhaustive. For campus demand, listing every microscopic weather and app-server state may be overspecified: it tracks details that do not change the decision.

Kolmogorov’s 1933 axioms (fundamental assumptions that define how probability must behave) made probability a theory of event sets, not just gambling arithmetic. This is why “what is an event?” matters.

Tier 2: probability space

Definition 1: A **probability space** is a triple $(\Omega, \mathcal{A}, P)$ where Ω is the sample space, $\mathcal{A}$ is the **event family** (a collection of subsets of Ω that we assign probabilities to; we write $\mathcal{A}$ to represent this family of events), and $P : \mathcal{A} \rightarrow [0, 1]$ (meaning “ P is a function that takes events from $\mathcal{A}$ as input and outputs numbers between 0 and 1”) satisfies:¹

$$P(A) \geq 0, \quad P(\Omega) = 1, \quad P(\cup_i A_i) = \sum_i P(A_i)$$

for countably many² disjoint events A_i (disjoint means no outcome belongs to more than one event—they don’t overlap), where $\bigcup_i A_i$ means “the union of all events $A_1, A_2, A_3, \dots$ ” (all outcomes that belong to at least one A_i).

For finite Ω , $\mathcal{A}$ is often the power set of Ω . For continuous spaces, such as the needle-in-a-square

¹In a fully formal treatment, $\mathcal{A}$ has additional technical conditions; we defer those conditions here.

²**Countably many** means either finitely many or indexed by natural numbers $i = 1, 2, 3, \dots$. In practice, this includes all finite and many infinite cases you’ll encounter.

example $\Omega = \{(x, y) \in \mathbb{R}^2 : 0 \leq x \leq 1, 0 \leq y \leq 1\}$, individual points have probability 0.³ Why? Because there are infinitely many points, and if each had positive probability, the total would exceed 1. To ask meaningful probability questions, we work with events that are regions (e.g., “does the needle land in the left half?”).

Conditional probability restricts attention to outcomes where some event B occurred, then renormalizes (rescales so probabilities sum to 1): $P(A|B) = \frac{P(A \cap B)}{P(B)}$ for $P(B) > 0$. Intuitively, $P(A|B)$ asks “what fraction of B -worlds also satisfy A ?” (Full treatment in Section 4.)

Lemma 1: For a fixed event B with $P(B) > 0$, the conditional probability function $P(\cdot | B)$ is itself a probability function on events.

Proof: Nonnegativity follows from $P(A \cap B) \geq 0$. Normalization is $P(\Omega | B) = \frac{P(B)}{P(B)} = 1$.

If A_i are disjoint, then $(A_i \cap B)$ are disjoint, so additivity follows after dividing by $P(B)$. $\square$

The probability-space definition now lets us compare what two plausible sample spaces can express.

Tier 3: comparing two sample spaces

Consider two models of one delivery:

$$\Omega_1 = \{\text{late, on time}\},$$

$$\Omega_2 = \{(\text{late, rain}), (\text{on time, rain}), (\text{late, dry}), (\text{on time, dry})\}.$$

Both are exhaustive and mutually exclusive. The question is which events their event families can contain.

In Ω_1 , the event “late” is $\{\text{late}\}$, but “rain” cannot be expressed because weather is absent from every outcome. In Ω_2 ,

$$\{(\text{late, rain}), (\text{on time, rain})\}$$

expresses “rain,” and

$$\{(\text{late, rain}), (\text{late, dry})\}$$

expresses “late.” The refined space can also express “late in rain” as the singleton $\{(\text{late, rain})\}$. Tier 2’s event family $\mathcal{A}$ therefore records the questions the representation makes available.

The coarser space is sufficient for a punctuality penalty. The refined space is needed to compare punctuality across weather. More detail is useful only when it supports a question or decision.

³This intuition assumes a non-atomic continuous model; mixed distributions can also have point masses.

Practice: Compare representations for one decision

Choose campus shuttle arrival, library seat availability, or phone battery status. Then produce a four-row design brief:

Row	Required output
1	State the decision the model should support, such as dispatch more shuttle buses, open overflow study rooms, or send a battery warning.
2	Give a coarse sample space with 2–4 mutually exclusive outcomes.
3	Give a refined sample space that adds one extra factor, such as weather, time period, or usage intensity.
4	Write one event that both spaces can express and one event that only the refined space can express. State which space supports the decision.

Optional stretch: name one variable that should be a random variable rather than part of the sample space.

Capability Checkpoint: Probability spaces and granularity

Work the numbered questions Q1–Q3 in the end-of-sheet **Checkpoint Question Bank**. They are tiered as Easy, Medium, and Hard; try them before reading the solutions.

2. Random variables and induced distributions

Outcomes describe full possible worlds, but decisions usually need simpler quantities. The lunch team does not inspect every detail of a day before choosing counters; it asks for quantities such as $P(D \geq 80)$, $P(L > 5)$, or expected workload. A random variable extracts such a quantity from each possible state, and its value-probabilities are what make these questions calculable.

Tier 1: random variables are functions

A random variable is a function that maps each outcome to a number or category. We write ω for a single outcome from the sample space Ω . For a lunch-day outcome ω , the notation $D(\omega)$ means “apply function D to outcome ω ”; for example:

- $D(\omega)$: number of lunch orders in outcome ω ;
- $L(\omega)$: number of late deliveries in outcome ω ;
- $T(\omega)$: delay time in minutes for a chosen order;
- $Y(\omega)$: early noisy app demand signal;
- $C(\omega)$: complaint count.

Tier 2: induced laws and preimages

How does a random variable get its probabilities? It **induces** (inherits or derives) them from the underlying outcome space: $P(X = x)$ equals the total probability of all outcomes ω where $X(\omega) = x$.

Read the next block as a definition: it names the function, its preimages, and the distribution inherited from Ω . The example afterward shows how to calculate with these objects in a finite case.

A random variable is “random” only because its input outcome is unknown. Mathematically it is an ordinary function with a probabilistic input.

Definition 2: A **random variable** is a function from outcomes to a value space, with preimages that are events. In this sheet we use both categorical variables, such as $W \in \{\text{dry}, \text{rain}\}$, and real-valued variables, such as $D \in \mathbb{R}$. Categorical variables are described by PMFs; real-valued variables may additionally have PDFs.

Before using this notation, note that $X^{\{-1\}}(S)$ is a **preimage**: the collection of all outcomes whose X -value lies in S (not an inverse function).

Its **induced distribution** is

$$P_X(S) = P(X \in S) = P(X^{-1}(S)) = P(\{\omega \in \Omega : X(\omega) \in S\}).$$

This formal definition is abstract. To see how it works in practice, consider a concrete case:

Example 1: Induced distribution. Let each outcome be a (weather, late-count) pair, and $\omega_1 = (\text{dry}, 0)$, $\omega_2 = (\text{dry}, 1)$, $\omega_3 = (\text{rain}, 1)$, with probabilities $P(\{\omega_1\}) = 0.5$, $P(\{\omega_2\}) = P(\{\omega_3\}) = 0.25$. Define L as the number of late orders, so $L(\omega_1) = 0$ and $L(\omega_2) = L(\omega_3) = 1$. Since “ $P(L = 0)$ ” is shorthand for $P(\{\omega : L(\omega) = 0\})$, we get $P(L = 0) = P(\{\omega_1\}) = 0.5$ and $P(L = 1) = P(\{\omega_2, \omega_3\}) = 0.5$. Thus L pushes the mass of Ω onto its values, merging ω_2, ω_3 into $L = 1$.

In the lunch model, the event “prepare for high demand” is the preimage

$$\{\omega : D(\omega) \geq 80\}.$$

If $P(\{\omega : D(\omega) \geq 80\}) = 0.30$, then the 30% comes from adding probabilities of all outcome-days where demand reaches 80.

The induced-distribution formula shows exactly which information a random variable retains and which distinctions it merges.

Tier 3: what an induced distribution keeps and loses

Can two different random variables have the same distribution? Yes. The distribution records the probabilities of values, not every feature of the outcomes that produced them.

Let $\Omega = \{\text{HH}, \text{HT}, \text{TH}, \text{TT}\}$ with equal probabilities. Define $X = 1$ when the first toss is heads and 0 otherwise. Define $Y = 1$ when the two tosses match and 0 otherwise. Here $X = 1$ on $\{\text{HH}, \text{HT}\}$, while $Y = 1$ on $\{\text{HH}, \text{TT}\}$. Using Tier 2's formula,

$$P_X(\{1\}) = P(\{\text{HH}, \text{HT}\}) = \frac{1}{2},$$

$$P_Y(\{1\}) = P(\{\text{HH}, \text{TT}\}) = \frac{1}{2}.$$

Both X and Y therefore have the Bernoulli($\frac{1}{2}$) distribution, although their preimages of $\{1\}$ are different.

The common distribution keeps $P(X = 0)$ and $P(X = 1)$, which is enough for questions about the value alone. It does not preserve whether the value records the first toss or agreement between the tosses. The map from outcomes to values supplies that missing meaning.

Practice: Construct equal distributions from different functions

On the same four-outcome space, define two new indicator variables with different preimages and the same Bernoulli parameter. List each preimage of $\{1\}$, use $P_Z(S) = P(Z^{-1}(S))$ to compute both induced distributions, and state one event about the original tosses that cannot be recovered from either distribution alone.

Capability Checkpoint: Random variables and induced distributions

Work the numbered questions Q4–Q6 in the end-of-sheet **Checkpoint Question Bank**. They are tiered as Easy, Medium, and Hard; try them before reading the solutions.

3. Discrete distributions and PMFs

Once random variables have values, their probability rules split by the kind of value. Demand category, late/not-late, complaint count, pickup counter, and “first late order” are **discrete** (they take values you can list out, either finitely many or countably infinite), so this section handles them with PMFs. Section 5 will handle measured quantities on a continuum.

Tier 1: probability functions for discrete values

A **discrete** random variable takes finite or countably many values (countable = can be listed as $v_1, v_2, v_3, \dots$). To work with a discrete random variable X , we need to know how probability is assigned to its values. The **probability mass function (PMF)** is the function $p_X(x) = P(X = x)$ that gives the probability for each value x . The complete assignment of probabilities is called the **distribution** of X .

Common patterns you’ll see:

- Bernoulli: one success/failure trial, e.g. “order is late.”
- Discrete uniform: each listed option is equally likely, e.g. a fair die.
- Binomial: number of successes in n independent Bernoulli trials.
- Geometric: number of trials until the first success.

Tier 2: formal definitions and specific distributions

Definition 3: A **PMF** of a discrete random variable X is the function $p_X(x) = P(X = x)$ with $p_X(x) \geq 0$ and $\sum_x p_X(x) = 1$. The **support** (the set of values that can actually occur) is $\{x : p_X(x) > 0\}$.

The four patterns above have these formal PMF definitions:

- Bernoulli(p): $p_X(1) = p$, $p_X(0) = 1 - p$. The indicator $\mathbf{1}_A$ is Bernoulli with parameter $P(A)$. For example, if A is the event “rain,” then $\mathbf{1}_A = 1$ when it rains and $\mathbf{1}_A = 0$ otherwise.
- Discrete uniform on $\{a, a + 1, \dots, b\}$: $p_X(i) = \frac{1}{b-a+1}$.
- Binomial(n, p): $p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}$ for $k = 0, \dots, n$, where $\binom{n}{k}$ is the number of ways to choose k items from n . It requires independent trials with common success probability.
- Geometric(p), counting trials until first success: $p_X(k) = (1-p)^{k-1} p$ for $k = 1, 2, \dots$, where $(1-p)^{k-1}$ represents $k-1$ failures before the first success.

The distributions above describe single random variables in isolation. But operational questions often involve **relationships** between variables: does demand increase when it rains? Does the number of late orders depend on the day of the week? To capture dependence, we need to model multiple random variables together.

When two discrete random variables X and Y can influence each other, their individual PMFs $p_X(x)$ and $p_Y(y)$ are not enough—we need their joint behavior:

Definition 4: The **joint PMF** of X, Y is $p_{X,Y}(x, y) = P(X = x, Y = y)$, giving the probability that both events happen simultaneously. The **marginals** and **conditionals** are

$$p_X(x) = \sum_y p_{X,Y}(x, y), \quad p_Y(y) = \sum_x p_{X,Y}(x, y),$$

$$p_{X|Y}(x|y) = \frac{p_{X,Y}(x, y)}{p_Y(y)} \quad \text{for } p_Y(y) > 0.$$

Definition 5: Two random variables X and Y are **independent**, written $X \perp Y$, if and only if

$$p_{X,Y}(x,y) = p_X(x)p_Y(y) \quad \text{for all } x,y.$$

When independence holds, knowing $Y = y$ tells you nothing about X : the conditional distribution $p_{X|Y}(x|y) = p_X(x)$ is the same as the marginal. Equivalently, $p_{Y|X}(y|x) = p_Y(y)$. In the lunch model, if weather and day-of-week are independent, then observing “it’s raining” doesn’t change your belief about whether it’s Monday or Friday. If demand and weather are **not** independent, then observing rain does change your demand forecast—this dependence is exactly what joint distributions capture.

Example 2: Campus table. Let D_c be demand category in $\{1,2,3\}$ and W be weather in $\{\text{dry}, \text{rain}\}$, with joint probabilities:

$$p(1, \text{dry}) = 0.20, p(2, \text{dry}) = 0.25, p(3, \text{dry}) = 0.15, p(1, \text{rain}) = 0.05, p(2, \text{rain}) = 0.15, p(3, \text{rain}) = 0.20.$$

Then $P(W = \text{rain}) = 0.40$, $P(D_c = 3) = 0.35$, and $P(D_c = 3 \mid W = \text{rain}) = \frac{0.20}{0.40} = 0.50$.

The joint-PMF definition lets us compare independent and dependent Bernoulli trials that have identical one-trial probabilities.

Tier 3: same marginals, different counts

Let I_1, I_2 be independent Bernoulli($\frac{1}{2}$) variables and set $N = I_1 + I_2$. Tier 2’s binomial formula gives

$$P(N = 0) = \frac{1}{4}, \quad P(N = 1) = \frac{1}{2}, \quad P(N = 2) = \frac{1}{4}.$$

Now let $B \sim \text{Bernoulli}(\frac{1}{2})$ and set $J_1 = B$, $J_2 = B$, so the trials always agree. Each J_i still has marginal probabilities $P(J_i = 1) = P(J_i = 0) = \frac{1}{2}$, but for $M = J_1 + J_2$,

$$P(M = 0) = \frac{1}{2}, \quad P(M = 1) = 0, \quad P(M = 2) = \frac{1}{2}.$$

The marginal Bernoulli probabilities do not determine the count distribution. Independence is the additional assumption that produces the binomial PMF; making successes and failures occur together moves probability toward the two extreme counts in this example.

Practice: Compare two Bernoulli-trial models

For two trials with $P(I_1 = 1) = P(I_2 = 1) = 0.2$, construct (a) an independent joint PMF and (b) a dependent joint PMF with the same marginals. Compute the PMF of the count in each model and identify which count is binomial.

Capability Checkpoint: Discrete distributions and PMFs

Work the numbered questions Q7–Q9 in the end-of-sheet **Checkpoint Question Bank**. They are tiered as Easy, Medium, and Hard; try them before reading the solutions.

4. Conditioning, product/sum rules, and Bayes

Conditioning is not only a story about time passing. It means changing the reference set or using partial information: among rainy days, among high-signal days, or among explanations that could have produced the observed app signal. This section links tables, densities, and hidden-state explanations through the same renormalization logic.

A useful update pipeline is:

- **prior**: what you believe before the new observation;
- **observation** or **evidence**: the data point or event you just saw;
- **likelihood**: how compatible each candidate explanation is with that observation;
- **evidence**: the overall probability of the observation across all explanations, used to normalize;
- **posterior**: the updated belief after conditioning on the observation.

Here, the **likelihood** is a function of the candidate explanation evaluated at the observed data, while the **evidence** is the normalizing total across all explanations. The **prior** and **posterior** are probability distributions before and after the update.

Bayes' rule gives three steps: **prior** -> **observation** -> **posterior**. If more data arrives later, the posterior from one step becomes the prior for the next step: prior -> observation_1 -> posterior_1 -> observation_2 -> posterior_2.

Tier 1: conditioning changes the reference population

Conditioning means: restrict attention to worlds where the evidence occurred, then renormalize. In the delivery example, $P(\text{late}|\text{rain})$ is not “late and rain” but “late among rainy cases.” This is the intuition behind the formula $P(A|B) = \frac{P(A \cap B)}{P(B)}$ from Section 1.

A Bayes update is the same idea with hidden causes: start from a **prior** belief about each cause, see an **observation** or **evidence** y , score each cause by its **likelihood** $P(y|\text{cause})$ or $f(y|\text{cause})$, divide by the total **evidence** $P(y)$ or $f(y)$, and obtain the **posterior**. In symbols,

$$\text{prior} \times \frac{\text{likelihood}}{\text{evidence}} = \text{posterior}$$

If new evidence arrives later, repeat the same pipeline: the posterior after one observation becomes the prior for the next observation. That is why Bayesian updating naturally forms a chain of continuations rather than a one-shot calculation.

Tier 2: rules for updating and marginalizing

Definition 6: For events A, B with $P(B) > 0$,

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

Base-rate neglect (ignoring how common the cause is before seeing evidence) is a recurring human error: people often focus on likelihood and forget the prior frequency of the cause.

Theorem 1: Product rule and chain rule.

$$p_{X,Y}(x, y) = p_X(x)p_{Y|X}(y|x) = p_Y(y)p_{X|Y}(x|y).$$

More generally, for d variables $X_1, \dots, X_d$:

$$p(x_1, \dots, x_d) = p(x_1)p(x_2|x_1)\dots p(x_d|x_1, \dots, x_{d-1}).$$

Here p is shorthand for the appropriate joint or conditional distribution at each step.⁴

Theorem 2: Sum rule / marginalization. The marginal distribution of X can be recovered from the joint distribution $p_{X,Y}$ by summing (or integrating) over all values of Y :

Discrete: $p_X(x) = \sum_y p_{X,Y}(x,y)$. Continuous: $f_X(x) = \int f_{X,Y}(x,y) dy$.

This shows that the joint distribution contains complete information about each individual variable's distribution.

Theorem 3: Bayes' rule. If X is the cause/state and Y is the result/noisy measurement, then

$$p_{X|Y}(x|y) = \frac{p_X(x)p_{Y|X}(y|x)}{p_Y(y)}, \quad p_Y(y) = \sum_{x'} p_X(x')p_{Y|X}(y|x')$$

in the discrete case, and

$$f_{X|Y}(x|y) = \frac{f_X(x)f_{Y|X}(y|x)}{f_Y(y)}, \quad f_Y(y) = \int f_X(x')f_{Y|X}(y|x') dx'$$

in the continuous case.

Example 3: Mixed case (discrete-continuous Bayes). Let $K \in \{0,1\}$ denote “no promotion/promotion” and Y be continuous measured demand. After observing $Y = y$, we are not just moving forward in time; we are comparing which hidden explanation makes that observation more plausible:

$$p_{K|Y}(k|y) = \frac{p_K(k)f_{Y|K}(y|k)}{\sum_{k'} p_K(k')f_{Y|K}(y|k')}.$$

The denominator is evidence: all explanations of the observed demand measurement.

Bayes' rule also shows separately how the prior and likelihood change the posterior.

Tier 3: how the prior and likelihood affect the posterior

For two states, **odds** convert probability p to $\frac{p}{1-p}$. Bayes' rule can then be written

$$\text{posterior odds} = \text{prior odds} \times \text{likelihood ratio},$$

where the likelihood ratio compares how likely the same observation is under the two states.

Let $K = 1$ mean promotion and $K = 0$ mean no promotion. At an observed signal y , suppose

$$f_{Y|K}(y|1) = 0.08, \quad f_{Y|K}(y|0) = 0.02.$$

The likelihood ratio is $\frac{0.08}{0.02} = 4$. Holding these likelihoods fixed:

Prior	Prior odds	Posterior odds	Posterior
0.10	$\frac{1}{9}$	$\frac{4}{9}$	$\frac{4}{13} \approx 0.308$
0.30	$\frac{3}{7}$	$\frac{12}{7}$	$\frac{12}{19} \approx 0.632$
0.60	$\frac{3}{2}$	6	$\frac{6}{7} \approx 0.857$

⁴The conditional factors are used only at conditioning values where they are defined. For continuous variables, conditional densities are defined up to probability-zero exceptions.

The observation multiplies the odds by the same factor of 4 in every row, but the posterior still depends on the prior. Likelihood evidence changes belief; it does not erase the starting prevalence.

Practice: Posterior sensitivity table

Keep likelihoods 0.08 and 0.02 fixed. Add rows for priors 0.05, 0.50, and 0.80, compute prior odds, posterior odds, and posterior probability, then state how a fixed likelihood ratio acts on different priors.

Check one row using the Tier 2 denominator formula:

Row	Required output
1	Write $P(K = 1)f(y K = 1) + P(K = 0)f(y K = 0)$.
2	Divide the promotion term by that evidence and compare it with the odds result.

Capability Checkpoint: Conditioning and Bayes

Work the numbered questions Q10–Q12 in the end-of-sheet **Checkpoint Question Bank**. They are tiered as Easy, Medium, and Hard; try them before reading the solutions.

5. Continuous distributions, PDFs, and CDFs

Section 3 handled listed or countable values. The other half of the Week 1 split is measurement on a continuum: delay time T , early app signal Y , and satisfaction score S . For these variables, probability lives in intervals and regions, not at exact points.

Tier 1: density is not point probability

For continuous delay time T , $P(T = 3.1) = 0$ under ordinary density models, but $P(3 \leq T \leq 4)$ can be positive. Why? The real line is **uncountable**—you cannot list all real numbers in any interval. Unlike discrete distributions where we sum over countably many values, continuous distributions assign probability to intervals, not points. A single point has zero “width” on the real line, so it carries zero probability. A PDF gives the probability density at a point: the rate at which probability accumulates in an infinitesimal interval around that point.

Tier 2: PDF, CDF, CCDF, joint laws

Read this section in two stages: the PDF, CDF, and joint density blocks define the objects; the integration, differentiation, and marginalization formulas are calculation rules for using them.

Definition 7: A continuous random variable X has PDF f_X if

$$P(X \in B) = \int_B f_X(x) dx, \quad f_X(x) \geq 0, \quad \int_{-\infty}^{\infty} f_X(x) dx = 1,$$

where $\int_B$ means “integrate over all x in region B ”.

A PDF can be larger than 1. This surprises many learners: density is NOT probability. Only area must be between 0 and 1. Example: uniform on $[0, 0.1]$ has density height 10, but the area (probability) is $10 \times 0.1 = 1$.

Interval probabilities are $P(a \leq X \leq b) = \int_a^b f_X(x) dx$. A density may be greater than 1; areas, not heights, are probabilities.

Definition 8: The **CDF** is $F_X(x) = P(X \leq x)$. For discrete variables, $F_X(x) = \sum_{k \leq x} p_X(k)$. For continuous variables, $F_X(x) = \int_{-\infty}^x f_X(t) dt$. The **CCDF** (complementary CDF) is $P(X > x) = 1 - F_X(x)$ for every x .

CDF properties: nondecreasing (never goes down as x increases), right-continuous (no downward jumps when approaching from the right), tends to 0 as $x \rightarrow -\infty$, and tends to 1 as $x \rightarrow \infty$.

For continuous random variables with differentiable CDF, the PDF and CDF are related by differentiation and integration: $f_X(x) = \frac{dF_X(x)}{dx}$ and $F_X(x) = \int_{-\infty}^x f_X(t) dt$. This makes them dual representations of the same distribution.

Definition 9: Jointly continuous variables have joint PDF $f_{X,Y}$ satisfying

$$P((X, Y) \in B) = \int \int_B f_{X,Y}(x, y) dx dy.$$

Marginals and conditionals are

$$f_X(x) = \int f_{X,Y}(x, y) dy, \quad f_{X|Y}(x|y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

when $f_Y(y) > 0$.

The joint CDF is $F_{X,Y}(x, y) = P(X \leq x, Y \leq y)$ and, when differentiable,

$$f_{X,Y}(x,y) = \frac{\partial^2 F_{X,Y}(x,y)}{\partial x \partial y},$$

where $\frac{\partial^2}{\partial x \partial y}$ means “differentiate once with respect to each variable” (requires calculus).

The PDF and CDF formulas provide two representations of the same interval probability.

Tier 3: one probability from a PDF and a CDF

Density warning. If $f_T(5) = 0.3$, it does not mean the probability of exactly 5 minutes delay is 0.3. It means a short interval around 5 minutes has probability approximately $0.3 \times$ interval length.

Let T have density $f_T(t) = 2t$ for $0 \leq t \leq 1$. Integrating from Tier 2 gives the CDF $F_T(t) = t^2$ on this interval. Now compute the same probability two ways:

$$P(0.5 \leq T \leq 1) = \int_{0.5}^1 2t \, dt = 1 - 0.25 = 0.75,$$

$$P(0.5 \leq T \leq 1) = F_T(1) - F_T(0.5) = 1 - 0.25 = 0.75.$$

The PDF accumulates local density across the interval; the CDF has already accumulated it up to each endpoint. They agree because the CDF is the integral of the PDF.

Practice: Verify a tail probability two ways

For the same T , compute $P(T > 0.6)$ by integrating the PDF and by using $1 - F_T(0.6)$. Then explain in one sentence why $f_T(0.6)$ is not either answer.

Capability Checkpoint: Continuous distributions, PDFs, and CDFs

Work the numbered questions Q13–Q15 in the end-of-sheet **Checkpoint Question Bank**. They are tiered as Easy, Medium, and Hard; try them before reading the solutions.

6. Expectation, variance, and indicator variables

Distributions tell you what values are possible and how likely they are. Many calculations need numerical summaries such as average delay or variability in demand. This section introduces expectation (probability-weighted average), variance (spread around the mean), and indicator variables (numerical representations of events).

Tier 1: expectation is a probability-weighted average

If you roll a fair die, the outcomes $\{1, 2, 3, 4, 5, 6\}$ each occur with probability $\frac{1}{6}$. The average outcome over many rolls is

$$1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} = 3.5.$$

This is the **expectation** of the die roll: the long-run average value, weighted by probability.

For the campus lunch team, if delivery delay T has PDF $f_T(t)$, the expected delay is not just “pick a typical value”—it’s the probability-weighted integral over all possible delays. Expectation answers: if we repeated this process many times, what would the average outcome be?

Tier 2: definitions and properties

Definition 10: The **expectation** (or **expected value** or **mean**) of a function $g(X)$ is:

Discrete: $E[g(X)] = \sum_x g(x)p_X(x)$, summing over all possible values x .

Continuous: $E[g(X)] = \int_{-\infty}^{\infty} g(x)f_X(x) dx$.

When $g(X) = X$ itself, we get the mean $\mu = E[X]$. When $g(X) = (X - \mu)^2$, we get variance (defined below).

For jointly distributed variables,

$$E[g(X, Y)] = \begin{cases} \sum_x \sum_y g(x, y)p_{X, Y}(x, y) & \text{(discrete)} \\ \int \int g(x, y)f_{X, Y}(x, y) dx dy & \text{(continuous)} \end{cases}$$

Theorem 4: Linearity of expectation. Assume X and Y are integrable, meaning $E[|X|] < \infty$ and $E[|Y|] < \infty$. Then, for any constants a, b, c ,

$$E[aX + bY + c] = aE[X] + bE[Y] + c.$$

This holds **regardless of whether X and Y are independent**. Expectation is always linear.

Definition 11: The **variance** of X measures spread around the mean:

$$\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2.$$

The **standard deviation** is $\sigma_X = \sqrt{\text{Var}(X)}$.

Properties:

- $\text{Var}(aX + b) = a^2 \text{Var}(X)$ (shifting by b doesn’t change spread; scaling by a scales variance by a^2).
- $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$, where covariance is defined below.

Definition 12: An **indicator variable** $\mathbf{1}_A$ equals 1 if event A occurs and 0 otherwise. Key property:

$$E[\mathbf{1}_A] = 1 \cdot P(A) + 0 \cdot P(\overline{A}) = P(A).$$

The indicator variable $\mathbf{1}_A$ has exactly the Bernoulli($P(A)$) distribution.⁵

Indicators let you turn event probabilities into expectations. Example: if A is “order arrives late,” then let A_i be the event that the i -th delivery is late. Then the expected number of late orders in n deliveries is $E[1_{A_1} + \dots + 1_{A_n}]$, and by linearity this equals $E[1_{A_1}] + \dots + E[1_{A_n}] = P(A_1) + \dots + P(A_n)$.

Definition 13: The **covariance** of X and Y measures how they vary together:

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])] = E[XY] - E[X]E[Y].$$

Properties:

- If X and Y are independent, then $E[XY] = E[X]E[Y]$, so $\text{Cov}(X, Y) = 0$.
- Positive covariance: X and Y tend to increase together.
- Negative covariance: when X increases, Y tends to decrease.
- Zero covariance does not imply independence (see margin note in Section 9).

Theorem 5: Law of total expectation (tower property). If X is integrable, then for any random variable Y ,

$$E[X] = E[E[X | Y]].$$

This says: to compute the overall average of X , you can first average within each value of Y , then average those conditional averages over all values of Y .

Discrete version: $E[X] = \sum_y E[X | Y = y]P(Y = y)$.

Continuous version: $E[X] = \int E[X | Y = y]f_Y(y) dy$.

Example 4: Let D be demand category (1, 2, or 3) and T be delivery delay time. If $E[T | D = 1] = 5$, $E[T | D = 2] = 8$, $E[T | D = 3] = 12$, and $P(D = 1) = 0.5$, $P(D = 2) = 0.3$, $P(D = 3) = 0.2$, then

$$E[T] = 5 \cdot 0.5 + 8 \cdot 0.3 + 12 \cdot 0.2 = 2.5 + 2.4 + 2.4 = 7.3.$$

Indicators and conditional expectation give two useful designs for the same expected-count calculation.

Tier 3: two routes to one expected count

Suppose there are 20 orders. Each order is late with probability 0.10 on a dry day and 0.30 on a rainy day, while $P(W = \text{rain}) = 0.40$. What is the expected number of late orders?

First use indicators. Let I_i indicate that order i is late and $N = \sum_{i=1}^{20} I_i$. The unconditional late probability is

The law of total expectation is also called the “tower property” because you can condition on a sequence of variables: $E[X] = E[E[X|Y]|Z]$. Each layer “averages out” one variable.

⁵The notation $X \stackrel{d}{=} Y$ means that X and Y are equal in distribution: they have the same probability distribution even though they need not be the same random variable.

$$P(I_i = 1) = 0.60 \cdot 0.10 + 0.40 \cdot 0.30 = 0.18.$$

By linearity,

$$E[N] = \sum_{i=1}^{20} E[I_i] = 20 \cdot 0.18 = 3.6.$$

Now condition on weather. The conditional expected counts are $E[N|W = \text{dry}] = 20 \cdot 0.10 = 2$ and $E[N|W = \text{rain}] = 20 \cdot 0.30 = 6$. The tower property gives

$$E[N] = 0.60 \cdot 2 + 0.40 \cdot 6 = 3.6.$$

The indicator route builds the total from orders; the conditional route builds it from weather groups. They agree because both are applications of Tier 2's expectation rules.

For variance, dependence still matters:

$$\text{Var}(N) = \sum_i \text{Var}(I_i) + 2 \sum_{i < j} \text{Cov}(I_i, I_j).$$

Orders that share the same weather can be positively related, so the expectation calculation alone does not justify dropping covariance.

Practice: Design two expectation calculations

Repeat both routes for 30 orders when $P(W = \text{rain}) = 0.25$, the dry-day late probability is 0.08, and the rainy-day late probability is 0.20. State why neither expected-count route requires the orders to be independent.

Capability Checkpoint: Expectation, variance, and indicators

Work the numbered questions Q16–Q18 in the end-of-sheet **Checkpoint Question Bank**. They are tiered as Easy, Medium, and Hard; try them before reading the solutions.

7. Gaussian toolkit

The Gaussian toolkit helps the lunch team model noisy measurements and heterogeneous days without inventing every formula from scratch. App measurement error may be many small effects; ordinary and promotion days may require mixtures; nonnegative, skewed delays warn us when Gaussian convenience is the wrong shape.

Tier 1: why Gaussian appears everywhere

The Gaussian is useful because total noise from combining many small independent or nearly-independent random effects is often modeled as Gaussian, and affine transformations of Gaussian variables remain Gaussian. Sums of independent Gaussian variables are Gaussian; more generally, any linear combination of a jointly Gaussian vector is Gaussian. In the campus model, noisy demand measurement might be approximated by $Y = D + \varepsilon$ with Gaussian measurement noise ε .

Tier 2: operational facts

Definition 14: We write $X \sim \mathcal{N}(0, 1)$ to mean “ X has a normal distribution with mean 0 and variance 1.”

Standard normal: $X \sim \mathcal{N}(0, 1)$ has

$$f_X(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right).$$

General normal: $X \sim \mathcal{N}(\mu, \sigma^2)$ has

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right).$$

Gaussian “normality” is partly justified by the Central Limit Theorem and partly chosen for mathematical convenience. Always ask: aggregation mechanism, measurement noise, or just convenience?

Example 5: Standardization. If $X \sim \mathcal{N}(\mu, \sigma^2)$, then $Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1)$.

Theorem 6: If $X \sim \mathcal{N}(\mu_X, \Sigma_X)$ with positive semidefinite covariance (a covariance matrix where all variances are non-negative—technical detail for multivariate cases) and $Y = AX + b$, then

$$Y \sim \mathcal{N}(A\mu_X + b, A\Sigma_X A^T).$$

If the covariance is singular, interpret this as a degenerate Gaussian on a lower-dimensional affine subspace (meaning the distribution is confined to a line or plane in higher-dimensional space—this is an edge case you can skip on first reading).

Σ_{XY} is cross-covariance (measures how X and Y vary together): in one dimension it is just $\text{Cov}(X, Y)$. In block matrices it records how components of X co-vary with components of Y .

Theorem 7: Marginals and conditionals of a jointly Gaussian vector are Gaussian. If the block covariance Σ_Y is invertible, then for

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}, \begin{pmatrix} \Sigma_X & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_Y \end{pmatrix}\right)$$

,

$$X|Y = y \sim \mathcal{N}(\mu_X + \Sigma_{XY}\Sigma_Y^{-1}(y - \mu_Y), \Sigma_X - \Sigma_{XY}\Sigma_Y^{-1}\Sigma_{YX}).$$

Lemma 2: The product of two Gaussian **densities** is proportional to a Gaussian density. If $f_i(x) = \mathcal{N}(x|\mu_i, \nu_i)$, then

$$f_0(x)f_1(x) \propto \mathcal{N}\left(x \mid \frac{\nu_1\mu_0 + \nu_0\mu_1}{\nu_0 + \nu_1}, \frac{\nu_0\nu_1}{\nu_0 + \nu_1}\right).$$

Distinguish two claims. “The product of Gaussian densities is Gaussian up to rescaling” is not the same as “the product XY of two Gaussian random variables is Gaussian.” The latter is generally false.

Example 6: Mixture of two Gaussian densities. If Z has density $f(z) = \alpha f_1(z) + (1 - \alpha)f_2(z)$ where f_i are $\mathcal{N}(\mu_i, \sigma_i^2)$ densities, then

$$E[Z] = \alpha\mu_1 + (1 - \alpha)\mu_2,$$

$$\text{Var}(Z) = \alpha\sigma_1^2 + (1 - \alpha)\sigma_2^2 + \alpha\mu_1^2 + (1 - \alpha)\mu_2^2 - (\alpha\mu_1 + (1 - \alpha)\mu_2)^2.$$

The affine and mixture formulas can be combined to build a two-group model and summarize its overall mean and variance.

Tier 3: building a model from Gaussian components

Let $B \sim \text{Bernoulli}(0.25)$ indicate a promotion day. Model demand X by

$$X|B = 0 \sim \mathcal{N}(70, 64), \quad X|B = 1 \sim \mathcal{N}(110, 144).$$

Suppose operational workload is $Y = 1.1X + 5$.

Tier 2’s affine rule gives

$$Y|B = 0 \sim \mathcal{N}(82, 77.44), \quad Y|B = 1 \sim \mathcal{N}(126, 174.24).$$

Using the mixture formula,

$$E[Y] = 0.75 \cdot 82 + 0.25 \cdot 126 = 93.$$

The within-group part of the variance is

$$0.75 \cdot 77.44 + 0.25 \cdot 174.24 = 101.64,$$

and the between-group part is

$$0.75(82 - 93)^2 + 0.25(126 - 93)^2 = 363.$$

Hence $\text{Var}(Y) = 464.64$.

A single Gaussian $\mathcal{N}(93, 464.64)$ matches these two summaries, but it represents one population. The mixture also represents ordinary and promotion regimes, including their different centers and spreads.

Practice: Construct and summarize a two-group model

Choose a group probability and two Gaussian component models, then apply an affine transformation $Y = aX + b$. Compute both conditional distributions, the mixture mean, the within-group variance, and the between-group variance. State what the mixture represents that a single Gaussian with the same mean and variance does not.

Capability Checkpoint: Gaussian toolkit

Work the numbered questions Q19–Q21 in the end-of-sheet **Checkpoint Question Bank**. They are tiered as Easy, Medium, and Hard; try them before reading the solutions.

8. Transformations and inverse-transform sampling

After building probability rules, the lunch team often needs derived quantities: convert delay into a penalty, demand into workload, or a fitted CDF into simulated scenarios before choosing staff. Transformations turn raw variables into operational quantities, and inverse-transform sampling provides a principled way to generate random samples for verification and stress-testing.

Tier 1: transform distributions through events

To find the distribution of $Y = g(X)$, start with events: $F_Y(y) = P(Y \leq y) = P(g(X) \leq y)$. Then differentiate if a PDF exists.

Example 7: If $f_X(x) = 3x^2$ for $0 \leq x \leq 1$ and $Y = X^2$, then for $0 \leq y \leq 1$,

$$F_Y(y) = P(X^2 \leq y) = P(X \leq \sqrt{y}) = \int_0^{\sqrt{y}} 3t^2 dt = y^{\frac{3}{2}},$$

so $f_Y(y) = \left(\frac{3}{2}\right)\sqrt{y}$.

Tier 2: change of variables and inverse sampling

In the next theorem, one-to-one differentiability is the assumption, the displayed density formula is the calculation rule, and the proof explains why that rule works.

Theorem 8: Change of variables. If $Y = U(X)$ where U is differentiable and one-to-one, then

$$f_Y(y) = f_X(U^{-1}(y)) \left| \frac{d}{dy} U^{-1}(y) \right|.$$

Intuition: Probability is conserved under transformation, so $f_Y(y) dy = f_X(x) dx$ where $x = U^{-1}(y)$. The absolute derivative $\left| \frac{dx}{dy} \right|$ accounts for how the transformation stretches or compresses intervals.

Proof: Assume U is strictly increasing. Then $F_Y(y) = P(Y \leq y) = P(U(X) \leq y) = P(X \leq U^{-1}(y)) = F_X(U^{-1}(y))$. Differentiate with respect to y using the chain rule: $f_Y(y) = f_X(U^{-1}(y)) \cdot \frac{dU^{-1}(y)}{dy}$. For decreasing U , the event $U(X) \leq y$ becomes $X \geq U^{-1}(y)$, so $F_Y(y) = 1 - F_X(U^{-1}(y))$, and differentiating yields the negative derivative. The absolute value handles both cases.

For non-one-to-one transformations, split into branches. For $Y = X^2$, values $x = \sqrt{y}$ and $x = -\sqrt{y}$ may both contribute if the support contains both.

Theorem 9: Probability integral transform. If X has continuous strictly increasing CDF F , then $U = F(X)$ is uniform on $[0, 1]$. Conversely, if $U \sim \text{Uniform}(0, 1)$, then $F^{-1}(U)$ has CDF F . For non-strict CDFs, use the generalized inverse $F^{-1}(u) = \inf\{x : F(x) \geq u\}$.

Example 8: Exponential inverse transform. For $F(x) = 1 - \exp(-\lambda x)$ on $x \geq 0$, solve $u = 1 - \exp(-\lambda x)$:

$$x = -\frac{\ln(1-u)}{\lambda}.$$

Inverse transform sampling turns a uniform random number generator into a distribution-specific generator.

Thus $Y = -\frac{\ln(1-U)}{\lambda}$ is exponential with rate λ .

Simulation workflow. Once you derive a transformation formula, simulation provides verification before trusting it for decisions:

1. Use the formula to generate random samples (e.g., 10,000 exponential values)
2. Compute sample statistics (mean, variance) from the simulated data
3. Compare to theoretical values (for exponential: mean = $\frac{1}{\lambda}$, variance = $\frac{1}{\lambda^2}$)
4. If they match $\rightarrow$ formula and code are correct. If not $\rightarrow$ debug before using the model.

Example: The exponential inverse transform $Y = -\frac{\ln(1-U)}{\lambda}$ should produce samples with mean $\approx \frac{1}{\lambda}$ and variance $\approx \frac{1}{\lambda^2}$:

```
import numpy as np
rng = np.random.default_rng(1001)
u = rng.uniform(size=10_000)
lam = 2.0
samples = -np.log1p(-u) / lam
print(samples.mean(), samples.var()) # about 1/lam, 1/lam**2
```

Simulation can check the CDF and density obtained from a Tier 2 transformation formula.

Tier 3: checking a transformation by simulation

Let X have density $f_X(x) = 3x^2$ on $[0, 1]$ and define $Y = X^2$. What should a simulation of Y reproduce?

Tier 2's CDF method gives, for $0 \leq y \leq 1$,

$$F_Y(y) = P(X^2 \leq y) = F_X(\sqrt{y}) = y^{\frac{3}{2}},$$

so $f_Y(y) = \frac{3}{2}\sqrt{y}$. It also gives

$$E[Y] = E[X^2] = \int_0^1 x^2(3x^2) dx = \frac{3}{5},$$

and $P(Y \leq 0.25) = 0.25^{\frac{3}{2}} = 0.125$.

Inverse-transform sampling uses $X = U^{\frac{1}{3}}$ for $U \sim \text{Uniform}(0, 1)$, followed by $Y = X^2$. A large simulation should therefore have mean near 0.6, about 12.5% of values at or below 0.25, and a histogram shaped like $3\frac{\sqrt{y}}{2}$.

Simulation does not prove the derivation, but agreement across a mean, a CDF value, and the density shape provides three checks of the same Tier 2 formula.

Practice: Simulate and compare

Generate at least 10,000 values using U , $X = U^{\frac{1}{3}}$, and $Y = X^2$. Report the sample mean, the sample fraction with $Y \leq 0.25$, and a histogram over $[0, 1]$. Compare all three with the theoretical results above.

Capability Checkpoint: Transformations and inverse-transform sampling

Work the numbered questions Q22–Q24 in the end-of-sheet **Checkpoint Question Bank**. They are tiered as Easy, Medium, and Hard; try them before reading the solutions.

9. Sums, convolution, and limit theorems

Real lunch operations aggregate: demand across counters, late deliveries across couriers, and complaints across service channels. Section 6 introduced expectation and variance for individual variables. This section asks what happens when you **add** random variables: what can be computed about the distribution of sums exactly (convolution), and what do the Law of Large Numbers and Central Limit Theorem tell us about averages of many variables?

Tier 1: sums aggregate uncertainty

Total lunch demand across locations is a sum. Total delay over a route is a sum. Sometimes the whole distribution is hard; mean and variance are easier.

Tier 2: convolution, summary statistics, and limit theorems

Theorem 10: If X and Y are independent and $Z = X + Y$, then

$$p_Z(z) = \sum_y p_X(z - y)p_Y(y)$$

in the discrete case, and

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(z - y)f_Y(y) dy$$

in the continuous case.

Example 9: Dice sum. Two fair dice produce a triangular PMF for $Z = X + Y$ because different sums have different numbers of contributing pairs. For example, $Z = 7$ can be formed by six pairs: $(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)$, while $Z = 2$ can only be formed by $(1, 1)$.

Theorem 11: Linearity of expectation does not require independence:

$$E[aX + bY + c] = aE[X] + bE[Y] + c.$$

Variance rules:

$$\text{Var}(aX + b) = a^2 \text{Var}(X),$$

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y).$$

If X and Y are independent, then $\text{Cov}(X, Y) = 0$ and variances add.

Law of Large Numbers

The Law of Large Numbers (LLN) formalizes the intuition that sample averages converge to the true mean as the sample size grows.

Independence is stronger than zero covariance. For example, let $X \sim \text{Uniform}(-1, 1)$ and $Y = X^2$. Then $\text{Cov}(X, Y) = 0$ by symmetry, but X and Y are clearly dependent. Variance additivity only needs zero covariance, but independence is a common sufficient condition.

Theorem 12: (Weak Law of Large Numbers) Let $X_1, X_2, \dots, X_n$ be i.i.d. (independent and identically distributed) random variables with finite mean $\mu = E[X_i]$ and finite variance $\sigma^2 = \text{Var}(X_i)$. Define the sample average

$$\bar{X}_n = \frac{X_1 + X_2 + \dots + X_n}{n}.$$

Then for any $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu| \geq \varepsilon) = 0.$$

Equivalently, $\bar{X}_n$ converges to μ in probability.

Operational meaning: If the lunch team records demand $D_1, D_2, \dots, D_n$ on n days (where each D_i is drawn i.i.d. from the same distribution with mean μ), then the sample average $\bar{D}_n = \frac{D_1 + \dots + D_n}{n}$ will be close to μ with high probability when n is large. The LLN justifies using long-run averages to estimate unknown population means.

Proof sketch. By linearity of expectation, $E[\bar{X}_n] = \mu$. By independence and variance rules,

$$\text{Var}(\bar{X}_n) = \text{Var}\left(\frac{X_1 + \dots + X_n}{n}\right) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}.$$

Applying Chebyshev's inequality,

$$P(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\text{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Example 10: Lunch demand estimation. Suppose daily demand D has mean $\mu = 75$ and variance $\sigma^2 = 100$. After observing $n = 25$ days, the sample average $\bar{D}_{25}$ has mean 75 and variance $\frac{100}{25} = 4$, so standard deviation 2. By Chebyshev, $P(|\bar{D}_{25} - 75| \geq 6) \leq \frac{4}{36} \approx 0.11$. As n grows, the sample average concentrates more tightly around the true mean.

Central Limit Theorem

The Central Limit Theorem (CLT) goes further than the LLN: it describes the **shape** of the distribution of sample averages, not just convergence to the mean.

Theorem 13: (Central Limit Theorem) Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with finite mean μ and finite variance $\sigma^2 > 0$. Define the standardized sum

$$Z_n = \frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}} = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}.$$

Then as $n \rightarrow \infty$, the distribution of Z_n converges to the standard normal distribution $\mathcal{N}(0, 1)$:

$$\lim_{n \rightarrow \infty} P(Z_n \leq z) = \Phi(z) = \int_{-\infty}^z \left(\frac{1}{\sqrt{2\pi}} \right) \exp\left(-\frac{t^2}{2}\right) dt.$$

Equivalently, $\bar{X}_n$ is approximately $\mathcal{N}(\mu, \sigma^2/n)$ for large n .

Operational meaning: Even if individual demands D_i are not Gaussian (e.g., discrete categories or bounded delays), the **average** demand $\bar{D}_n$ over many days behaves approximately like a Gaussian with mean μ and variance σ^2/n . This justifies Gaussian approximations for aggregated quantities and enables normal-based confidence intervals and hypothesis tests.

Example 11: Lunch delivery time aggregate. Assume $T_1, \dots, T_{36}$ are i.i.d. delivery delays, each with mean $\mu = 5$ minutes and variance $\sigma^2 = 9$. Their sum has mean $36 \times 5 = 180$ and variance $36 \times 9 = 324$, so standard deviation $\sqrt{324} = 18$. By CLT, the total delay $S_{36} = \sum_{i=1}^{36} T_i$ is approximately $\mathcal{N}(180, 324)$. Thus

$$P(S_{36} \leq 200) \approx \Phi\left(\frac{200 - 180}{18}\right) = \Phi(1.11) \approx 0.87.$$

The team can use this Gaussian approximation to set delivery time budgets even if individual T_i is not Gaussian.

Tier 2 offers exact, summary, and approximate views of the same sum; comparing them shows what each calculation provides.

Tier 3: exact and approximate calculations for a sum

Let $I_1, \dots, I_{40}$ be independent Bernoulli(0.3) variables and let $S = \sum_{i=1}^{40} I_i$. We want $P(S \geq 16)$.

The exact route repeatedly convolves the Bernoulli PMF; equivalently, Tier 2's binomial formula gives

$$P(S \geq 16) = \sum_{k=16}^{40} \binom{40}{k} 0.3^k 0.7^{40-k} \approx 0.115.$$

This calculation provides the complete discrete distribution and an exact tail probability.

The summary route gives

$$E[S] = 40 \cdot 0.3 = 12, \quad \text{Var}(S) = 40 \cdot 0.3 \cdot 0.7 = 8.4.$$

These numbers locate the sum and describe its spread, but they do not determine the requested tail by themselves.

The CLT route approximates S by $\mathcal{N}(12, 8.4)$. Using a continuity correction, which replaces the discrete threshold $S \geq 16$ by the continuous threshold 15.5,

$$P(S \geq 16) \approx 1 - \Phi\left(\frac{15.5 - 12}{\sqrt{8.4}}\right) \approx 0.114.$$

This route is quick and close here, while the exact binomial sum remains the reference. The approximation can be poor when the number of trials is small or the success probability is very close to 0 or 1.

Practice: Compare three calculations for one sum

Repeat the exact binomial tail, mean/variance, and continuity-corrected CLT approximation for $S \sim \text{Binomial}(60, 0.4)$ and the event $S \geq 30$. Explain in one sentence what each method contributes.

Capability Checkpoint: Sums, limit theorems, and dependence

Work the numbered questions Q25–Q27 in the end-of-sheet **Checkpoint Question Bank**. They are tiered as Easy, Medium, and Hard; try them before reading the solutions.

10. Entropy, KL divergence, and decisions

The final step is model comparison. Before deploying a staffing rule, the team may compare two demand-category models P and Q : does a model miss rare high-demand days, or place probability on the wrong regions? Entropy summarizes uncertainty; KL measures directed distributional mismatch through expected excess log-loss when one distribution is used while another generates the data.

Tier 1: compare uncertainty and mismatch

Entropy quantifies how spread out or concentrated a distribution is. A uniform distribution over many outcomes has high entropy (maximum unpredictability): “demand could be anything from 50 to 150 with equal likelihood.” A concentrated distribution has low entropy (high predictability): “demand is almost always between 70 and 80.” Entropy gives the ideal lower bound on how many optimally chosen yes/no questions you would need, on average, to pin down the outcome.

KL divergence measures the expected excess log-loss from using Q when data follow P . KL divergence $D_{\text{KL}}(P \parallel Q)$ quantifies this mismatch from P ’s perspective. An operational loss function can separately evaluate the cost of staffing decisions based on the forecast. In general, $D_{\text{KL}}(P \parallel Q) \neq D_{\text{KL}}(Q \parallel P)$ because the two directions use different reference distributions.

Entropy describes uncertainty, while cross-entropy and KL compare a predictive distribution with a reference distribution. An operational loss is still needed to choose an action.

Tier 2: definitions, properties, and binary example

Definition 15: For a discrete distribution p , Shannon entropy (named after Claude Shannon, who founded information theory) is

$$H(p) = - \sum_x p(x) \log p(x).$$

With base-2 logarithms, entropy is measured in bits. It is the ideal lower bound on the average number of optimally chosen yes/no questions needed to identify the outcome.⁶

Why entropy measures both uncertainty and information. A uniform distribution over many outcomes has high entropy because observing the outcome resolves substantial uncertainty. A distribution concentrated on one outcome has low entropy because the observation is usually predictable.

Example 12: Demand category entropy. Suppose demand category $D_c \in \{1, 2, 3\}$ follows two possible distributions:

- Uniform: $p_1(1) = p_1(2) = p_1(3) = \frac{1}{3}$, so $H(p_1) = -3 \times (\frac{1}{3}) \log_2(\frac{1}{3}) = \log_2(3) \approx 1.58$ bits.
- Concentrated: $p_2(1) = 0.1$, $p_2(2) = 0.8$, $p_2(3) = 0.1$, so $H(p_2) \approx 0.92$ bits.

The uniform distribution has higher entropy: demand is less predictable, but observing the outcome gives more information. The concentrated distribution has lower entropy: demand is more predictable (usually category 2), so observing the outcome gives less information.

⁶For an optimal one-outcome binary prefix code with expected length L , $H(X) \leq L < H(X) + 1$; equality is not always attainable. With long block codes, the average number of bits per outcome can approach $H(X)$.

Definition 16: The KL divergence $D_{\text{KL}}(P \parallel Q)$ measures the expected excess log-loss when data follow P but we use model Q :

$$D_{\text{KL}}(P \parallel Q) = \sum_x P(x) \log \left(\frac{P(x)}{Q(x)} \right)$$

in the discrete case, and

$$D_{\text{KL}}(P \parallel Q) = \int p(x) \log \left(\frac{p(x)}{q(x)} \right) dx$$

in the continuous case, when P is absolutely continuous with respect to Q (technical condition: roughly, P doesn't put probability where Q assigns zero).

The decomposition $D_{\text{KL}}(P \parallel Q) = H(P, Q) - H(P)$ separates KL divergence into two pieces. Here $H(P, Q) = -\sum_x P(x) \log Q(x)$ is the **cross-entropy**: the expected log-loss when outcomes follow P but predictions use Q , with an equivalent ideal code-length interpretation. The term $H(P)$ is the corresponding baseline when predictions use the true distribution P . Thus KL divergence is the expected excess log-loss from using model Q when the true distribution is P .

In the lunch model, if the true demand distribution is P but the team forecasts with Q , KL divergence diagnoses the average predictive mismatch. The team can then combine the forecast with an operational loss function that assigns costs to under- and overstaffing. This connects distributional mismatch to the staffing consequences that matter for the decision.

KL divergence has four key properties that distinguish it from ordinary distances:

- **Non-negativity:** $D_{\text{KL}}(P \parallel Q) \geq 0$, with equality only when $P = Q$ almost everywhere. Under data generated by P , predictions based on Q cannot have lower expected log-loss than predictions based on P .
- **Asymmetry:** $D_{\text{KL}}(P \parallel Q) \neq D_{\text{KL}}(Q \parallel P)$ in general. The expected excess log-loss from using Q when truth is P differs from that of using P when truth is Q . Direction matters.
- **Not a distance:** KL does not obey the triangle inequality $d(a, c) \leq d(a, b) + d(b, c)$ that distances satisfy. It is a directed divergence, not a metric.
- **Infinite penalty:** In the discrete case, if $P(x) > 0$ but $Q(x) = 0$ for some outcome x , then $D_{\text{KL}}(P \parallel Q) = \infty$. In the continuous or general case, the divergence is infinite if P assigns positive probability to a region that Q assigns zero probability.

Example 13: Binary KL with natural logs. Let $P(0) = 0.9$, $P(1) = 0.1$, $Q(0) = 0.5$, $Q(1) = 0.5$.

$$D_{\text{KL}}(P \parallel Q) = 0.9 \ln \left(\frac{0.9}{0.5} \right) + 0.1 \ln \left(\frac{0.1}{0.5} \right) \approx 0.368.$$

$$D_{\text{KL}}(Q \parallel P) = 0.5 \ln \left(\frac{0.5}{0.9} \right) + 0.5 \ln \left(\frac{0.5}{0.1} \right) \approx 0.511.$$

Using base-2 logs gives bits; using natural logs gives nats (natural units, using e as the logarithm base).

The cross-entropy decomposition can compare predictive models, while a separate expected-loss calculation evaluates a staffing action.

Tier 3: using KL to compare predictive models

Let the reference distribution of low, medium, and high demand be

$$P = (0.50, 0.30, 0.20).$$

Compare candidate forecasts

$$Q_A = (0.45, 0.35, 0.20), \quad Q_B = (0.60, 0.25, 0.15).$$

Using natural logarithms, Tier 2's formulas give

Forecast	Cross-entropy $H(P, Q)$	KL $D_{\text{KL}}(P \ Q)$
Q_A	1.036	0.006
Q_B	1.051	0.021

Both columns prefer Q_A . Their rankings agree because $H(P, Q) = H(P) + D_{\text{KL}}(P \| Q)$ and the same $H(P) \approx 1.030$ is added for both candidates. Cross-entropy is expected log-loss; KL is the expected excess over predicting with P itself.

Now evaluate actions with a separate operational loss. Suppose standard staffing costs 0, 1, 8 units under low, medium, and high demand, while expanded staffing costs 3, 1, 0. Using Q_A ,

$$E[\text{loss} | \text{standard}, Q_A] = 0.45(0) + 0.35(1) + 0.20(8) = 1.95,$$

$$E[\text{loss} | \text{expanded}, Q_A] = 0.45(3) + 0.35(1) + 0.20(0) = 1.70.$$

The forecast comparison says Q_A is closer to P under log-loss. The action calculation says expanded staffing has lower expected operational loss under Q_A . These are related through the forecast but answer different questions.

Practice: Separate predictive and operational comparisons

Compute cross-entropy and KL for a third candidate $Q_C = (0.50, 0.35, 0.15)$. Then use the same staffing-loss table to choose an action under Q_C . Report the predictive comparison and action comparison separately.

Capability Checkpoint: Entropy, KL, and decisions

Work the numbered questions Q28–Q30 in the end-of-sheet **Checkpoint Question Bank**. They are tiered as Easy, Medium, and Hard; try them before reading the solutions.

11. Closing synthesis and activities

Return to the opening question and the 11:30 lunch decision. A Week 1 modeling sequence is:

1. choose what possible states matter for the decision;
2. extract useful quantities from states, such as D , L , T , Y , and C ;
3. choose a probability rule appropriate to discrete values or continuous measurements;
4. condition to focus on a reference group or compare hidden explanations;
5. transform or simulate variables to answer operational “what if” questions;
6. aggregate workload and uncertainty, including dependence;
7. compare models and check whether they assign probability to all relevant outcomes.

Key Takeaway

Probability modeling requires a connected sequence of choices: define outcomes, extract useful quantities, assign probabilities, condition on information, transform or simulate variables, aggregate uncertainty, and compare predictions with observations and decisions.

PROJECT **Mini-project checkpoint.** For a campus or personal stochastic process, submit: sample space, at least two random variables, one PMF/PDF/CDF, one conditional or Bayes update, one simulation diagnostic, and one paragraph titled “What this model still gets wrong.”

AI **AI-assisted path.** Ask an AI tool for a model of campus food demand. Then examine it: identify its sample space, random variables, independence assumptions, likelihood/evidence terms, and one place where the proposed model is too simple.

NON-AI **Non-AI path.** Build the same model from a hand-drawn tree/table. Exchange with a peer and revise one assumption after peer critique.

Thought Experiment: Noisy measurement versus true demand

Suppose true demand D is low, medium, or high, but the app reports a noisy continuous measurement Y . Which is more natural: $p(D|Y)$, $f(Y|D)$, or both? What information would you need to compute each direction?

Annotated reading references

Deisenroth, Faisal, and Ong, *Mathematics for Machine Learning*

Recommended companion for probability, Gaussian models, change of variables, and machine-learning uses of KL divergence.

Week 1 coding habit

Simulate first simple processes, estimate empirical PMF/CDF, compare empirical and theoretical mean/variance, and write assumptions before trusting a model.

Checkpoint Question Bank and Solutions

Section 1: Probability spaces and granularity

Q1 Easy. A delivery is either late or not late. Propose a sample space.

Solution.

$$\{\text{late, on time}\} = \mathcal{U}$$

Q2 Medium. Add rain/no-rain to the delivery model. State the sample space and one event.

Solution.

One sample space is $\mathcal{U} = \{(\text{late, rain}), (\text{on time, rain}), (\text{late, dry}), (\text{on time, dry})\}$. The event “rain” is

Q3 Hard. A team proposes outcomes {low demand, high demand, rain} for predicting staffing. Critique and repair it.

Solution.

The outcomes mix demand states and a weather state, so they are not mutually exclusive: “high demand and rain” can both occur. A repair is a product sample space such as $\{(\text{low, rain}), (\text{low, no rain}), (\text{high, rain}), (\text{high, no rain})\}$, or a richer space with demand count and weather.

Section 2: Random variables and induced distributions

Q4 Easy. On two coin tosses, define X = number of heads. What is $X(\text{HT})$?

Solution.

$$1 = (\text{JH})X$$

Q5 Medium. For $\Omega = \{\text{HH}, \text{HT}, \text{TH}, \text{TT}\}$ with equal probabilities, compute $P_X(\{1, 2\})$ for X = number of heads.

Solution.

$$X \in \{1, 2\} \text{ occurs for HH, HT, and TH, so } P^X(\{1, 2\}) = \frac{3}{4}.$$

Q6 Hard. In a demand model, distinguish the outcome, event, random variable, value, and distribution for the claim “high demand happens with probability 0.3.”

Solution.

Outcome: one full possible state of the system, e.g. weather, promotion, and order count. Random variable: $D(\omega)$, demand count or demand category. Value: high. Event: $\{\omega : D(\omega) = \text{high}\}$. Distribution: P_D , with $P_D(\text{high}) = 0.3$.

Section 3: Discrete distributions and PMFs

Q7 Easy. If late-delivery indicator $I \sim \text{Bernoulli}(0.2)$, what are $P(I = 1)$ and $P(I = 0)$?

Solution.

$$P(I = 1) = 0.2 \text{ and } P(I = 0) = 0.8.$$

Q8 Medium. Let $X \in \{1, 2, 3\}$ denote demand category and let $Y \in \{\text{dry}, \text{rain}\}$ denote weather. Their joint PMF is $p(1, \text{dry}) = 0.20$, $p(2, \text{dry}) = 0.25$, $p(3, \text{dry}) = 0.15$, $p(1, \text{rain}) = 0.05$, $p(2, \text{rain}) = 0.15$, $p(3, \text{rain}) = 0.20$. Compute $P(X = 2)$ and $P(Y = \text{rain} | X = 2)$.

Solution.

$$P(X=2) = 0.25 + 0.15 = 0.40. P(Y=\min(X=2)) = 0.15 + 0.40 = 0.55.$$

Q9 Hard. A promotion day creates social contagion: one late order makes later orders more likely to be late. Should the number of late orders in 20 orders be binomial?

Solution.

Not exactly. Binomial requires independent trials with common success probability. Contagion creates dependence, and the probability of lateness changes after previous late orders. A binomial model could be used as a rough approximation only if this dependence is negligible or explicitly ignored.

Section 4: Conditioning and Bayes

Q10 Easy. If $P(A \cap B) = 0.12$ and $P(B) = 0.30$, compute $P(A|B)$.

Solution.

$$P(A|B) = \frac{0.12}{0.30} = 0.40.$$

Q11 Medium. Promotion prior is $P(K=1) = 0.3$. Demand measurement likelihoods are $f(y|K=1) = 0.08$ and $f(y|K=0) = 0.02$ at the observed y . Compute $P(K=1|Y=y)$.

Solution.

$$\text{Evidence is } 0.3 * 0.08 + 0.7 * 0.02 = 0.038. \text{ Posterior is } \frac{0.024}{0.038} \approx 0.632.$$

Q12 Hard. Explain why the evidence denominator in Bayes' rule must include every modeled explanation, not just the preferred one.

Solution.

The denominator sums or integrates over all modeled explanations for the observation. If an important alternative explanation is omitted, the posterior can be overconfident. Thus the denominator forces comparison among causes and exposes whether the model's hypothesis space is adequate.

Section 5: Continuous distributions, PDFs, and CDFs

Q13 Easy. If X is uniform on $[0, 0.1]$, what is its density height?

Solution.

$$\text{The density is 10 on } [0, 0.1], \text{ because } 10 * 0.1 = 1.$$

Q14 Medium. If $f_X(x) = 2x$ for $0 \leq x \leq 1$, compute $P(0.5 \leq X \leq 1)$.

Solution.

$$\int_{0.5}^1 2x \, dx = x^2 \Big|_{0.5}^1 = 1 - 0.25 = 0.75.$$

Q15 Hard. A joint density is constant on the unit square. Compute $P(X+Y \leq 1)$ and explain geometrically.

Solution.

The density is 1 on area 1. The event $X+Y \leq 1$ is the lower-left triangle of the unit square with area $\frac{1}{2}$, so the probability is $\frac{1}{2}$.

Section 6: Expectation, variance, and indicators

Q16 Easy. Let A be the event that a delivery is late, with $P(A) = 0.2$. What values can the indicator $\mathbf{1}_A$ take, and what is its expectation?

Solution.

The indicator $\mathbf{1}_A$ equals 1 when A occurs and 0 otherwise. Therefore $E[\mathbf{1}_A] = P(A) = 0.2$.

Q17 Medium. Suppose $E[X] = 3$, $E[Y] = 4$, $\text{Var}(X) = 2$, $\text{Var}(Y) = 5$, and $\text{Cov}(X, Y) = 1$. Compute $E[2X - Y + 5]$ and $\text{Var}(X + Y)$.

Solution.

By linearity, $E[2X - Y + 5] = 2 \cdot 3 - 4 + 5 = 7$. Using the covariance term, $\text{Var}(X + Y) = 2 + 5 + 2 \cdot 1 = 9$.

Q18 Hard. Let weather W satisfy $P(W = \text{rain}) = 0.3$. Suppose $E[T \mid W = \text{rain}] = 12$ minutes and $E[T \mid W = \text{dry}] = 6$ minutes. Compute $E[T]$ and explain whether this calculation requires T and W to be independent.

Solution.

By the law of total expectation, $E[T] = 12 \cdot 0.3 + 6 \cdot 0.7 = 7.8$ minutes. Independence is not required: the conditional means explicitly allow the distribution of T to differ between rainy and dry days.

Section 7: Gaussian toolkit

Q19 Easy. If $X \sim \mathcal{N}(10, 4)$, what is the distribution of $\frac{X-10}{2}$?

Solution.

It is $\mathcal{N}(0, 1)$.

Q20 Medium. If $X \sim \mathcal{N}(5, 9)$ and $Y = 2X + 1$, find the mean and variance of Y .

Solution.

$E[Y] = 2 \cdot 5 + 1 = 11$. $\text{Var}(Y) = 2^2 \cdot 9 = 36$. Thus $Y \sim \mathcal{N}(11, 36)$.

Q21 Hard. When would a mixture of two Gaussians be more defensible than a single Gaussian for campus demand?

Solution.

When observations plausibly come from two regimes, such as ordinary days and promotion days, or dry days and rainy days. A single Gaussian may place its mean between regimes and misrepresent both; a mixture represents within-regime spread and between-regime differences.

Section 8: Transformations and simulation

Q22 Easy. If $U \sim \text{Uniform}(0, 1)$ and $X = -\frac{\ln(1-U)}{2}$, what distribution does X have?

Solution.

Exponential with rate $\lambda = 2$.

Q23 Medium. If $f_X(x) = 3x^2$ on $[0, 1]$ and $Y = X^2$, compute $F_Y(y)$ for $0 \leq y \leq 1$.

Solution.

$F_Y(y) = P(X^2 \leq y) = P(X \leq \sqrt{y}) = \int_0^{\sqrt{y}} 3t^2 dt = \sqrt{y}^3 = y^{\frac{3}{2}}$.

Q24 Hard. Why can the one-to-one change-of-variable formula fail for $Y = X^2$ when X has support $[-1, 1]$?

Solution.

The map $x \mapsto x^2$ is not one-to-one on $[-1, 1]$: both $\sqrt{y}$ and $-\sqrt{y}$ map to y . The density must sum contributions from both branches, each with its derivative correction.

Section 9: Sums and summary statistics

Q25 Easy. If $E[X] = 3$ and $E[Y] = 4$, what is $E[2X - Y + 5]$?

Solution.

$$2 \cdot 3 - 4 + 5 = 7.$$

Q26 Medium. If independent X, Y have variances 2 and 5, compute $\text{Var}(3X + Y)$.

Solution.

$$\text{Var}(3X + Y) = 9 \text{Var}(X) + \text{Var}(Y) = 9 \cdot 2 + 5 = 23.$$

Q27 Hard. Rainy days increase both demand and delivery delay. Explain why adding variances may understate uncertainty in total workload.

Solution.

Demand and delay are positively dependent through rain, so covariance is positive. For a sum, variance includes $2\text{Cov}(X, Y)$. Ignoring this term treats variables as uncorrelated and can underestimate spread.

Section 10: Entropy, KL, and decisions

Q28 Easy. Which has larger entropy: a fair coin or a coin with $P(H) = 0.99$?

Solution.

The fair coin has larger entropy because outcomes are more uncertain.

Q29 Medium. Let $P = (0.8, 0.2)$ and $Q = (0.5, 0.5)$. Write the expression for $D_{\text{KL}}(P \| Q)$.

Solution.

$$D_{\text{KL}}(P \| Q) = 0.8 \log\left(\frac{0.8}{0.5}\right) + 0.2 \log\left(\frac{0.2}{0.5}\right).$$

Q30 Hard. Suppose model Q assigns zero probability to a rare but possible failure that model P assigns positive probability. What happens to $D_{\text{KL}}(P \| Q)$, and what modeling warning does this give?

Solution.

$D_{\text{KL}}(P \| Q) = \infty$ because the term with $P(x) > 0$ and $Q(x) = 0$ has infinite log ratio. A model that assigns zero probability to a possible outcome can therefore have infinite KL divergence even if it fits common cases.

Appendices: derivation bank

Appendix A. Gaussian conditional formula

For a joint Gaussian vector with invertible Σ_Y ,

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}, \begin{pmatrix} \Sigma_X & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_Y \end{pmatrix}\right),$$

conditioning on $Y = y$ gives

$$X|Y = y \sim \mathcal{N}(\mu_X + \Sigma_{XY}\Sigma_Y^{-1}(y - \mu_Y), \Sigma_X - \Sigma_{XY}\Sigma_Y^{-1}\Sigma_{YX}).$$

This formula is the algebra behind Gaussian regression and Kalman-style updates.

Appendix B. Product of Gaussian densities

For scalar variances ν_0, ν_1 ,

$$\begin{aligned} & \exp\left(-\frac{(x - \mu_0)^2}{2\nu_0}\right) \exp\left(-\frac{(x - \mu_1)^2}{2\nu_1}\right) \\ &= \exp\left(-\frac{1}{2}\left[\left(\frac{1}{\nu_0} + \frac{1}{\nu_1}\right)x^2 - 2\left(\frac{\mu_0}{\nu_0} + \frac{\mu_1}{\nu_1}\right)x + c\right]\right). \end{aligned}$$

Completing the square gives variance

$$\nu = \left(\frac{1}{\nu_0} + \frac{1}{\nu_1}\right)^{-1} = \nu_0 \frac{\nu_1}{\nu_0 + \nu_1}$$

and mean

$$\nu\left(\frac{\mu_0}{\nu_0} + \frac{\mu_1}{\nu_1}\right) = \frac{\nu_1\mu_0 + \nu_0\mu_1}{\nu_0 + \nu_1}.$$

The missing multiplicative constant is a normalization factor.

Appendix C. Mixture mean and variance

Let $B \sim \text{Bernoulli}(\alpha)$ and draw $Z|B = 1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$, $Z|B = 0 \sim \mathcal{N}(\mu_2, \sigma_2^2)$. Then

$$E[Z] = E[E[Z|B]] = \alpha\mu_1 + (1 - \alpha)\mu_2.$$

Also

$$E[Z^2] = \alpha(\sigma_1^2 + \mu_1^2) + (1 - \alpha)(\sigma_2^2 + \mu_2^2),$$

so $\text{Var}(Z) = E[Z^2] - E[Z]^2$, yielding the formula in the main text. The second part is between-component variance.

Appendix D. Multivariate change of variables

If $Y = U(X)$ is differentiable and invertible from $\mathbb{R}^d$ to $\mathbb{R}^d$, then

$$f_Y(y) = f_X(U^{-1}(y))|\det(J_{U^{-1}}(y))|.$$

For $Y = AX + b$ with invertible A , this becomes

$$f_Y(y) = f_X(A^{-1}(y - b))|\det(A^{-1})|.$$

For standard Gaussian X , the exponent becomes a quadratic form in y , producing the multivariate Gaussian affine-transform rule.

Appendix E. Entropy and KL derivation scaffold

Shannon's entropy is justified by three desiderata: continuity in probabilities, monotonic increase with the number of equally likely outcomes, and a grouping rule saying uncertainty from a two-stage choice should be the weighted sum of stage uncertainties. These conditions lead, up to a positive constant, to

$$H(p_1, \dots, p_n) = - \sum_i p_i \log p_i.$$

For KL,

$$\begin{aligned} D_{\text{KL}}(P \| Q) &= \sum_x P(x) \log \left(\frac{P(x)}{Q(x)} \right) \\ &= - \sum_x P(x) \log Q(x) - \left(- \sum_x P(x) \log P(x) \right) = H(P, Q) - H(P). \end{aligned}$$

Thus KL is excess cross-entropy relative to the entropy of the true source.